

Assignment 5 — C Programming Debugging: Errors & Corrections

Topic: Arrays, strings, and array-based statistics (total/average/highest/lowest). Each table lists the buggy line, error type, explanation, and the corrected code.

File: 1.c — Student Marks System

Erroneous Code	Error Type	Explanation	Corrected Code
<code>int marks[5] = {80,90,70,60,50}</code>	Syntax Error	Missing semicolon after array initialization.	<code>int marks[5] = {80,90,70,60,50};</code>
<code>for(int i=0;i<=5;i++) { total += marks[i]; }</code>	Array Bounds Error	<code>marks[5]</code> has valid indices 0-4; <code>i<=5</code> accesses <code>marks[5]</code> , which is out of bounds.	<code>for(int i=0;i<5;i++) { total += marks[i]; }</code>
<code>printf("Average = %d\n", average);</code>	Type Mismatch	<code>average</code> is declared as float but <code>%d</code> (int format) is used.	<code>printf("Average = %.2f\n", average);</code>
<code>scores[5] = 60;</code>	Array Bounds Error	<code>scores[5]</code> only has indices 0-4; index 5 is out of bounds.	Remove this line, or declare <code>scores[6]</code> if a 6th value is intended.
<code>char subject[5]; strcpy(subject, "Programming");</code>	Array Bounds / Buffer Overflow	"Programming" needs 12 bytes (11 chars + null terminator) but <code>subject[]</code> only holds 5.	<code>char subject[12]; strcpy(subject, "Programming");</code>
<code>attendance["one"] = 1;</code>	Type Mismatch / Syntax Error	Array index must be an integer; a string literal cannot be used as an index.	<code>attendance[0] = 1;</code>
<code>attendance[2.5] = 1;</code>	Type Mismatch	Array index must be an integer; a float index is invalid.	<code>attendance[2] = 1;</code>
<code>for(int i=0;i<=5;i++) { printf("%d ", attendance[i]); }</code>	Array Bounds Error	<code>attendance[5]</code> has valid indices 0-4; <code>i<=5</code> reads out of bounds.	<code>for(int i=0;i<5;i++) { printf("%d ", attendance[i]); }</code>
<code>scanf("%d", arr[i]);</code>	Runtime Error (Pointer Error)	<code>scanf</code> needs the address of the variable; <code>arr[i]</code> is a value, not a pointer.	<code>scanf("%d", &arr[i]);</code>
<code>grade[0] = "A";</code>	Type Mismatch	<code>grade[0]</code> is a single char; a double-quoted string cannot be assigned to it.	<code>grade[0] = 'A';</code>
<code>if(prices[2] = 300)</code>	Logical Error	Assignment '=' used instead of comparison '=='; this overwrites <code>prices[2]</code> instead of checking its value.	<code>if(prices[2] == 300)</code>
<code>inventory = prices;</code>	Syntax / Logical Error	Arrays cannot be assigned to each other directly with '='; array names are not modifiable lvalues.	<code>for(int i=0;i<5;i++) inventory[i] = prices[i]; // or use memcpy(inventory, prices, sizeof(prices));</code>
<code>printf("%d\n", salary[0]);</code>	Type Mismatch	<code>salary[]</code> is a float array but <code>%d</code> is used to print it.	<code>printf("%.2f\n", salary[0]);</code>
<code>city = "Jaipur";</code>	Syntax / Logical Error	A char array name cannot be assigned a string directly with '='.	<code>strcpy(city, "Jaipur");</code>

Erroneous Code	Error Type	Explanation	Corrected Code
<pre>for(int i=0;i<=5;i++) { result[i] = i*10; }</pre>	Array Bounds Error	result[5] has valid indices 0-4; loop should not include index 5.	<pre>for(int i=0;i<5;i++) { result[i] = i*10; }</pre>

File: 2.c — Array Operations Program

Erroneous Code	Error Type	Explanation	Corrected Code
<code>for(int i=0;i<=10;i++) { printf("%d ", numbers[i]); }</code>	Array Bounds Error	numbers[10] has valid indices 0-9; i<=10 reads numbers[10], out of bounds.	<code>for(int i=0;i<10;i++) { printf("%d ", numbers[i]); }</code>
<code>sum = numbers[i];</code>	Logical Error	Uses '=' instead of '+=', so sum just gets overwritten each loop instead of accumulating.	<code>sum += numbers[i];</code>
<code>printf("Average = %d\n", average);</code>	Type Mismatch	average is a float, but %d is used.	<code>printf("Average = %.2f\n", average);</code>
<code>if(marks[i] < highest) { highest = marks[i]; }</code>	Logical Error	Condition is inverted for finding the highest value; should check for greater than, not less than.	<code>if(marks[i] > highest) { highest = marks[i]; }</code>
<code>attendance[5] = 1;</code>	Array Bounds Error	attendance[5] array has valid indices 0-4; index 5 is out of bounds.	Remove this line, or declare attendance [6] if a 6th day is intended.
<code>for(int i=0;i<=5;i++) { present += attendance[i]; }</code>	Array Bounds Error	Same as above — loop reads one index past the array's bounds.	<code>for(int i=0;i<5;i++) { present += attendance[i]; }</code>
<code>char subject[10]; strcpy(subject,"ComputerScience");</code>	Buffer Overflow	"ComputerScience" needs 16 bytes but subject[] only holds 10.	<code>char subject[16]; strcpy(subject,"ComputerScience");</code>
<code>branch = "IT";</code>	Syntax / Logical Error	A char array cannot be assigned a string with '='.	<code>strcpy(branch, "IT");</code>
<code>fees["two"] = 50000;</code>	Type Mismatch	Array index must be an integer, not a string.	<code>fees[2] = 50000;</code>
<code>fees[3.5] = 60000;</code>	Type Mismatch	Array index must be an integer, not a float.	<code>fees[3] = 60000;</code>
<code>scanf("%d", result[i]);</code>	Runtime Error (Pointer Error)	scanf requires the address of the variable, not its value.	<code>scanf("%d", &result[i]);</code>
<code>if(stock[2] = 300)</code>	Logical Error	Assignment used instead of comparison.	<code>if(stock[2] == 300)</code>
<code>backup = stock;</code>	Syntax / Logical Error	Arrays cannot be assigned with '='.	<code>for(int i=0;i<5;i++) backup[i] = stock[i];</code>
<code>printf("%d\n", salary[0]);</code>	Type Mismatch	salary[] is a float array; %d should be %f.	<code>printf("%.2f\n", salary[0]);</code>
<code>for(int i=0;i<=5;i++) { inventory[i] = i; }</code>	Array Bounds Error	inventory[5] has valid indices 0-4; loop should not include index 5.	<code>for(int i=0;i<5;i++) { inventory[i] = i; }</code>
<code>char city[8]; strcpy(city,"JaipurCity");</code>	Buffer Overflow	"JaipurCity" needs 11 bytes but city[] only holds 8.	<code>char city[11]; strcpy(city,"JaipurCity");</code>
<code>count = inventory;</code>	Syntax / Logical Error	Arrays cannot be assigned directly with '='.	<code>for(int i=0;i<5;i++) count[i] = inventory[i];</code>
<code>printf("Program Ended\n")</code>	Syntax Error	Missing semicolon at end of statement.	<code>printf("Program Ended\n");</code>

File: 3.c — Student Result Analysis

Erroneous Code	Error Type	Explanation	Corrected Code
<code>if(marks[i] > lowest) { lowest = marks[i]; }</code>	Logical Error	Condition is inverted for finding the lowest value; should check for less than.	<code>if(marks[i] < lowest) { lowest = marks[i]; }</code>
<code>for(int i=0;i<=5;i++) { present += attendance[i]; }</code>	Array Bounds Error	attendance[5] has valid indices 0-4; loop reads one past the end.	<code>for(int i=0;i<5;i++) { present += attendance[i]; }</code>
<code>char subject[8]; strcpy(subject, "Programming");</code>	Buffer Overflow	"Programming" needs 12 bytes but subject[] only holds 8.	<code>char subject[12]; strcpy(subject, "Programming");</code>
<code>scanf("%s",&branch);</code>	Logical Error (not strictly invalid, but incorrect style)	For %s, the array name itself already decays to a pointer; using &branch; passes the wrong pointer type (works only by coincidence on arrays).	<code>scanf("%s", branch); // no & needed for char arrays</code>
<code>fees[5] = 100000;</code>	Array Bounds Error	fees[5] array has valid indices 0-4; index 5 is out of bounds.	Remove this line, or declare fees[6] if a 6th value is intended.
<code>if(stock[2] = 30)</code>	Logical Error	Assignment used instead of comparison.	<code>if(stock[2] == 30)</code>
<code>backup = stock;</code>	Syntax / Logical Error	Arrays cannot be assigned with '='.	<code>for(int i=0;i<5;i++) backup[i] = stock[i];</code>
<code>printf("%d\n", salary[0]);</code>	Type Mismatch	salary[] is a float array; should use %f.	<code>printf("%.2f\n", salary[0]);</code>
<code>scanf("%d", result[i]);</code>	Runtime Error (Pointer Error)	Missing & before result[i]; scanf needs the address.	<code>scanf("%d", &result[i]);</code>
<code>city = "Jaipur";</code>	Syntax / Logical Error	A char array name cannot be assigned a string with '='.	<code>strcpy(city, "Jaipur");</code>
<code>inventory["one"] = 100;</code>	Type Mismatch	Array index must be an integer, not a string.	<code>inventory[0] = 100;</code>
<code>inventory[2.5] = 200;</code>	Type Mismatch	Array index must be an integer, not a float.	<code>inventory[2] = 200;</code>
<code>for(int i=0;i<=5;i++) { printf("%d ", inventory[i]); }</code>	Array Bounds Error	Loop should stop before index 5 (valid range 0-4).	<code>for(int i=0;i<5;i++) { printf("%d ", inventory[i]); }</code>
<code>count = inventory;</code>	Syntax / Logical Error	Arrays cannot be assigned with '='.	<code>for(int i=0;i<5;i++) count[i] = inventory[i];</code>
<code>char course[6]; strcpy(course, "Computer");</code>	Buffer Overflow	"Computer" needs 9 bytes but course[] only holds 6.	<code>char course[9]; strcpy(course, "Computer");</code>
<code>for(int i=0;i<=5;i++) { numbers[i] = i*10; }</code>	Array Bounds Error	numbers[5] has valid indices 0-4; loop should not reach index 5.	<code>for(int i=0;i<5;i++) { numbers[i] = i*10; }</code>
<code>return 0</code>	Syntax Error	Missing semicolon at the end of the return statement.	<code>return 0;</code>

File: 4.c — Sales Report System

Erroneous Code	Error Type	Explanation	Corrected Code
<code>printf("Average Sales = %d\n", averageSales);</code>	Type Mismatch	averageSales is a float but %d is used.	<code>printf("Average Sales = %.2f\n", averageSales);</code>
<code>if(sales[i] < highestSale) { highestSale = sales[i]; }</code>	Logical Error	Condition is inverted for finding the highest value.	<code>if(sales[i] > highestSale) { highestSale = sales[i]; }</code>
<code>monthlySales[5] = 250;</code>	Array Bounds Error	monthlySales[5] has valid indices 0-4; index 5 is out of bounds.	Remove this line, or declare monthlySales[6].
<code>for(int i=0;i<=5;i++) { printf("%d ", monthlySales[i]); }</code>	Array Bounds Error	Loop should stop before index 5.	<code>for(int i=0;i<5;i++) { printf("%d ", monthlySales[i]); }</code>
<code>char product[8]; strcpy(product, "Laptop Computer");</code>	Buffer Overflow	"LaptopComputer" needs 15 bytes but product[] only holds 8.	<code>char product[15]; strcpy(product, "LaptopComputer");</code>
<code>category = "Electronics";</code>	Syntax / Logical Error	A char array cannot be assigned a string with '='.	<code>strcpy(category, "Electronics");</code>
<code>stock["first"] = 100;</code>	Type Mismatch	Array index must be an integer, not a string.	<code>stock[0] = 100;</code>
<code>stock[2.7] = 200;</code>	Type Mismatch	Array index must be an integer, not a float.	<code>stock[2] = 200;</code>
<code>scanf("%d", orders[i]);</code>	Runtime Error (Pointer Error)	Missing & before orders[i].	<code>scanf("%d", &orders[i]);</code>
<code>if(revenue[2] = 3000)</code>	Logical Error	Assignment used instead of comparison.	<code>if(revenue[2] == 3000)</code>
<code>backupRevenue = revenue;</code>	Syntax / Logical Error	Arrays cannot be assigned with '='.	<code>for(int i=0;i<5;i++) backupRevenue[i] = revenue[i];</code>
<code>printf("%d\n", commission[0]);</code>	Type Mismatch	commission[] is a float array; should use %f.	<code>printf("%.2f\n", commission[0]);</code>
<code>city = "Jaipur";</code>	Syntax / Logical Error	Char array cannot be assigned a string with '='.	<code>strcpy(city, "Jaipur");</code>
<code>char branch[5]; branch[0..3] set, no null terminator</code>	Logical Error (Missing Null Terminator)	branch[] is filled with 4 characters but never null-terminated, so %s reads past the array (undefined behaviour).	<code>branch[4] = '\0'; // add null terminator before printf</code>
<code>for(int i=0;i<=5;i++) { employees[i] = i*100; }</code>	Array Bounds Error	Loop should stop before index 5.	<code>for(int i=0;i<5;i++) { employees[i] = i*100; }</code>
<code>char department[6]; strcpy(department, "Marketing");</code>	Buffer Overflow	"Marketing" needs 10 bytes but department[] only holds 6.	<code>char department[10]; strcpy(department, "Marketing");</code>
<code>expenses = sales;</code>	Syntax / Logical Error	Arrays cannot be assigned with '='.	<code>for(int i=0;i<5;i++) expenses[i] = sales[i];</code>
<code>ratings[-1] = 5;</code>	Array Bounds Error	Negative index -1 is out of bounds and writes to memory before the array (undefined behaviour).	<code>ratings[0] = 5; // use a valid index 0-4</code>

Erroneous Code	Error Type	Explanation	Corrected Code
<code>printf("%s\n", feedback[0]);</code>	Type Mismatch	feedback[0] is an int but %s (string) is used; this is undefined behaviour.	<code>printf("%d\n", feedback[0]);</code>
<code>printf("Report Generated Successfully\n")</code>	Syntax Error	Missing semicolon at end of statement.	<code>printf("Report Generated Successfully\n");</code>

File: 5.c — Student Performance Analysis

Erroneous Code	Error Type	Explanation	Corrected Code
<code>printf("Average = %d\n", average);</code>	Type Mismatch	average is a float but %d is used.	<code>printf("Average = %.2f\n", average);</code>
<code>if(marks[i] < highest) { highest = marks[i]; }</code>	Logical Error	Condition inverted for finding the highest value.	<code>if(marks[i] > highest) { highest = marks[i]; }</code>
<code>if(marks[i] > lowest) { lowest = marks[i]; }</code>	Logical Error	Condition inverted for finding the lowest value.	<code>if(marks[i] < lowest) { lowest = marks[i]; }</code>
<code>attendance[10] = 0;</code>	Array Bounds Error	attendance[10] array has valid indices 0-9; index 10 is out of bounds.	Remove this line, or declare attendance [11].
<code>for(int i=0;i<=10;i++) { present += attendance[i]; }</code>	Array Bounds Error	Loop should stop before index 10 (valid range 0-9).	<code>for(int i=0;i<10;i++) { present += attendance[i]; }</code>
<code>char department[20]; department = "InformationTechnology";</code>	Syntax / Logical Error	A char array cannot be assigned a string with '='; also "InformationTechnology" (22 chars) exceeds department[20].	<code>char department[23]; strcpy(department, "InformationTechnology");</code>
<code>fees["two"] = 50000;</code>	Type Mismatch	Array index must be an integer, not a string.	<code>fees[2] = 50000;</code>
<code>fees[4.5] = 70000;</code>	Type Mismatch	Array index must be an integer, not a float.	<code>fees[4] = 70000;</code>
<code>scanf("%d", result[i]);</code>	Runtime Error (Pointer Error)	Missing & before result[i].	<code>scanf("%d", &result[i]);</code>
<code>if(inventory[2] = 30)</code>	Logical Error	Assignment used instead of comparison.	<code>if(inventory[2] == 30)</code>
<code>backup = inventory;</code>	Syntax / Logical Error	Arrays cannot be assigned with '='.	<code>for(int i=0;i<5;i++) backup[i] = inventory[i];</code>
<code>printf("%d\n", salary[0]);</code>	Type Mismatch	salary[] is a float array; should use %f.	<code>printf("%.2f\n", salary[0]);</code>
<code>city = "Jaipur";</code>	Syntax / Logical Error	Char array cannot be assigned a string with '='.	<code>strcpy(city, "Jaipur");</code>
<code>char branch[5]; branch[0..3] set, no null terminator</code>	Logical Error (Missing Null Terminator)	Only 4 characters are set with no '\0', so %s reads past the array bounds.	<code>branch[4] = '\0'; // add null terminator before printf</code>
<code>for(int i=0;i<=5;i++) { stock[i] = i*100; }</code>	Array Bounds Error	stock[5] has valid indices 0-4; loop should not reach index 5.	<code>for(int i=0;i<5;i++) { stock[i] = i*100; }</code>
<code>ratings[-1] = 10;</code>	Array Bounds Error	Negative index is out of bounds (undefined behaviour).	<code>ratings[0] = 10;</code>
<code>printf("%s\n", feedback[0]);</code>	Type Mismatch	feedback[0] is an int but %s is used.	<code>printf("%d\n", feedback[0]);</code>
<code>scores = marks;</code>	Syntax / Logical Error	Arrays cannot be assigned with '='.	<code>for(int i=0;i<5;i++) scores[i] = marks[i];</code>

Erroneous Code	Error Type	Explanation	Corrected Code
<pre>printf("%d\n", temp[5]);</pre>	Array Bounds Error	temp[5] array has valid indices 0-4; temp[5] is out of bounds.	<pre>printf("%d\n", temp[4]); // use a valid index 0-4</pre>